

Development of Physically Constrained Machine Learning Surrogates for Rapid Settlement Prediction in Complex Tropical Peats

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DEVELOPMENT OF PHYSICALLY CONSTRAINED MACHINE LEARNING SURROGATES FOR RAPID SETTLEMENT PREDICTION IN COMPLEX TROPICAL PEATS

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Abstract

Tropical peat deposits in Malaysia, comprising fibric, hemic, and sapric materials, are extremely compressible as the water content often >200% and reported up to 2200%, initial void ratio typically 5 up to 30, and low shear strength of from 5 to 25 kPa, creating major risks for road embankments through excessive consolidation settlement and instability. This study aimed to quantify when classical one-dimensional (1D) consolidation theory remains valid and when two-dimensional (2D) stress redistribution demands advanced modelling, and to develop physically constrained machine-learning surrogates Artificial Neural Network (ANN) versus Support Vector Regression (SVR) for rapid settlement prediction. Settlement behavior was first examined through a parametric sensitivity framework comparing a theoretical 1D analytical baseline with 2D finite element analysis using the Soft Soil Model, revealing close alignment for decomposed peats under standard loading but divergence up to 21% for highly fibrous peats at maximum void ratios, indicating thresholds where lateral yielding and multidirectional stress transfer dominate. A synthetic database was then generated by varying governing embankment-peat variables across realistic Malaysian ranges, producing 24,000 scenarios; physical feasibility was enforced via a settlement-to-depth constraint ($S_T/D_p < 1.0$), removing around 3,000 non-physical cases with including extreme case of $cc = 3.15$ with $e_o = 15$ at shallow depths and yielding 21,000 high-fidelity samples. An MLP ANN (9 inputs; 128–64 hidden neurons; tanh; Adam; $L2 = 0.0001$; early stopping) and an RBF-kernel SVR (GridSearchCV, 5-fold CV, standardized inputs) were trained, validated and tested (70/15/15) and evaluated using R^2 , RMSE, and MAE. Both achieved $R^2 > 0.999$. As per prediction, SVR provided superior global accuracy on testing ($R^2 = 0.999872$, RMSE = 8.033 mm, MAE = 6.025 mm) versus ANN ($R^2 = 0.999655$, RMSE = 13.190 mm, MAE = 9.646 mm). Correlation and trend verification confirmed settlement is primarily governed by peat depth ($r = 0.55$) and compression index ($r = 0.53$), with controlled inverse sensitivity to e_o and weaker but physically consistent effects of groundwater level and embankment loading. Overall, SVR is recommended for high-precision prediction within the trained domain, while ANN showed stronger physical fidelity in isolated parametric interpretations, underscoring that geotechnical ML surrogates should be judged by both statistical performance and mechanistic consistency.

Keywords: Tropical peat, Finite element modelling, Settlement prediction, Soft Soil Model, Parametric analysis, Machine learning surrogates.

1 Introduction

Tropical peat deposits are among the most problematic geomaterials encountered in infrastructure development across Malaysia and other humid equatorial regions. These deposits are commonly characterized by extremely high natural water content, very large initial void ratio, high organic content, low unit weight, and very low undrained shear strength [1], [2], [3], [4], [5]. Depending on the degree of decomposition, Malaysian peat may be classified as fibric, hemic, or sapric, each exhibiting different compressibility and deformation characteristics [6], [7]. Fibric peat generally contains a more open fibre structure and tends to experience larger deformation, whereas sapric peat is more decomposed and comparatively less compressible [8], [9], [10], [11]. These highly variable properties create major challenges for road embankment construction, particularly in the form of excessive consolidation settlement, long-term creep deformation, lateral spreading, and potential instability [12], [13], [14], [15], [16], [17], [18].

Settlement prediction over peat remains a difficult task because the deformation response is governed by coupled soil structure, compressibility, groundwater condition, stress history, and embankment loading [12], [13], [17], [18], [19], [20]. Classical one-dimensional consolidation theory is widely used in preliminary geotechnical design because of its simplicity and clear physical basis [21], [22], [23], [24]. However, embankment loading over soft peat rarely produces a purely one-dimensional strain condition [16], [25], [26]. In the field, stress redistribution, lateral yielding, and multidirectional deformation may occur, particularly in highly fibrous peat or under high embankment surcharge [27], [28], [29], [30]. As a result, one-dimensional analytical solutions may provide reasonable estimates for some conditions but may become less reliable when the peat deposit exhibits very high compressibility or when the applied loading causes significant two-dimensional deformation [31], [32], [33], [34].

Classical one-dimensional consolidation theory is widely used in preliminary geotechnical design because of its simplicity and clear physical basis [21], [22], [23], [24]. However, embankment loading over soft peat rarely produces a purely one-dimensional strain condition [16], [25], [26]. In the field, stress redistribution, lateral yielding, and multidirectional deformation may occur, particularly in highly fibrous peat or under high embankment surcharge [27], [28], [29], [30]. As a result, one-dimensional analytical solutions may provide reasonable estimates for some conditions but may become less reliable when the peat deposit exhibits very high compressibility or when the applied loading causes significant two-dimensional deformation [31], [32], [33], [34].

Finite element modelling provides a more advanced framework for capturing the nonlinear stress-strain response and spatial deformation mechanism of peat under embankment loading [16], [18], [35]. Constitutive models such as the Soft Soil Model are particularly useful for representing the compressible behaviour of normally consolidated and lightly overconsolidated soft soils [36], [37]. Nevertheless, finite element analysis requires careful parameter calibration, mesh development, boundary condition definition, and computational effort [38], [39]. These requirements limit its direct use for rapid design screening, sensitivity analysis, or repeated evaluation of large parametric combinations. Therefore, there is a practical need for surrogate models that can retain the predictive strength of advanced numerical analysis while enabling fast settlement estimation [40], [41], [42].

Machine learning (ML) methods have increasingly been applied in geotechnical engineering because of their ability to capture nonlinear relationships between input parameters and engineering responses [43], [44], [45]. Artificial Neural Networks (ANN) and Support Vector Regression (SVR) are particularly suitable for settlement prediction because they can approximate complex nonlinear functions without requiring an explicit closed-form equation [46], [47], [48]. However, the use of machine learning in geotechnical design must be approached carefully. A model with high statistical accuracy is not necessarily reliable for engineering interpretation if it fails to reproduce physically meaningful trends [49], [50]. For peat settlement problems, a surrogate model should not only minimize prediction error, but also respect fundamental soil mechanics principles, such as the influence of peat thickness, compression index, void ratio, groundwater level, and surcharge stress [51], [52], [53], [54].

In this study, an integrated analytical numerical machine learning framework was developed for settlement prediction of embankments over Malaysian tropical peat. The study first examined the relationship between classical one-dimensional consolidation theory and two-dimensional finite element analysis using the Soft Soil Model (SSM). This comparison was used to identify the conditions under which the simplified analytical approach remains reliable and the conditions under which two-dimensional stress redistribution becomes significant. A large synthetic database was then generated by varying key embankment and peat parameters across realistic Malaysian ranges. To ensure physical reliability, the database was refined using a settlement-to-depth constraint, where non-physical cases with settlement exceeding peat thickness were removed before model training.

The refined database was used to train and evaluate two machine learning surrogates: an optimized Multi-Layer Perceptron ANN and a Radial Basis Function SVR model. Both models were assessed using global statistical indicators, including R^2 , RMSE, and MAE. In addition, parametric trend verification was performed to determine whether the models could reproduce expected geotechnical behaviour when individual input variables were varied independently. This dual evaluation is important because it distinguishes between statistical prediction accuracy and physical consistency, which is essential for engineering design applications.

The main contribution of this study is the development of a physically constrained machine learning framework for rapid settlement prediction over tropical peat. Unlike purely data-driven approaches, the proposed framework integrates analytical consolidation theory, finite element interpretation, physical feasibility filtering, and parametric verification. This allows the ANN and SVR models to be evaluated not only as numerical predictors, but also as geotechnical surrogates capable of supporting practical design decisions. The findings provide insight into the dominant settlement drivers in Malaysian peat and clarify the relative suitability of ANN and SVR for high-accuracy prediction and physically consistent parametric interpretation.

2 Methodology

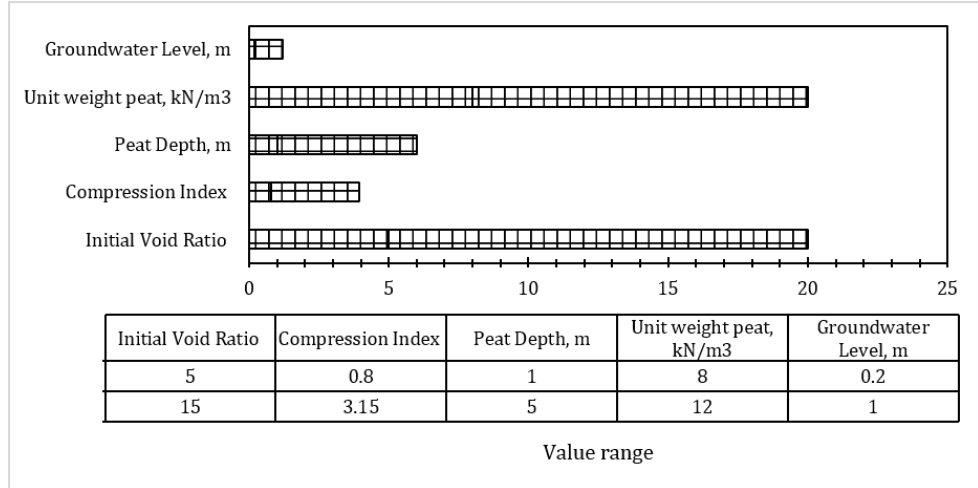
The research methodology is structured as an integrated computational pipeline, transitioning from raw data synthesis to physically constrained refinement, and concluding with a comparative ML performance audit.

2.1 Synthetic Database Generation

The development of a robust ML predictive model requires a comprehensive and representative training database. Given the high heterogeneity of Malaysian tropical peat influenced by varying degrees of decomposition (fibric, hemic, and sapric) and botanical origins a secondary database was compiled to establish realistic boundary

conditions for synthetic data generation. This database systematically integrates geotechnical parameters from peer-reviewed literature, regional soil investigations, and technical reports to ensure that the ML training set reflects the unique mechanical behaviour of ultra-soft organic soils. To simulate the non-linear consolidation characteristics of peatlands, a parametric matrix was designed by varying eight critical governing variables identified in the initial sensitivity analysis based on the equation from 1 and 2. Thus, a total of 24,000 unique scenarios were generated using these established ranges, summarized below which includes compressibility parameters as shown in Fig. 1. This extensive parametric range ensures that the ANN and SVR architectures are exposed to a wide spectrum of field conditions. By utilizing this multi-variate approach, the resulting database provides sufficient data density to accurately map the non-linear relationship between index properties and total settlement (S_T), bridging the gap between simplified 1D analytical theory and high-fidelity 2D numerical simulations.

Fig. 1 Geotechnical parameter ranges for Malaysian tropical peat used in the parametric modelling framework [2], [6], [10], [11], [28], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64].



These parameter ranges were used to generate both analytical settlement calculations and numerical modelling scenarios, allowing for a consistent comparison between theoretical predictions and finite element results.

$$S_T = \frac{c_c D_p}{1+e_o} \log \left(\frac{\sigma'_o + \Delta\sigma}{\sigma'_o} \right) \quad (1)$$

where e_o is the initial void ratio, σ'_o is the initial effective vertical stress, and $\Delta\sigma$ is the applied stress increment of the embankment.

The observed trends are consistent with the classical one-dimensional consolidation relationship:

$$S_T = \frac{c_c D_p}{1+e_o} \log \frac{\sigma'_o + \Delta\sigma_z}{\sigma'_o} + C'_\alpha D_p \log \left(\frac{t_1}{t_2} \right) \quad (2)$$

where S_T is total settlement, D_p is the thickness of the compressible peat layer, c_c is the compression index, e_o is the initial void ratio, σ'_o is the initial effective vertical stress, $\Delta\sigma$ is the applied stress increment, C'_α is secondary compression index which from fraction of C_c , typically $0.06 C_c$, t_1 is time at the end of primary consolidation (year) and t_2 is target time for total settlement (year)

2.2 Data Refinement and Physical Constraints

In the application of machine learning to geotechnical engineering, the integrity and physical validity of the training dataset are paramount to ensuring model reliability. While the initial 24,000 scenarios generated through the 1D analytical framework provided a robust mathematical foundation, purely theoretical datasets often include computational artifacts results that are numerically consistent with formulas but physically unattainable in a field environment. To maintain Geotechnical Fidelity, a rigorous Physical Feasibility Filter was applied to the raw data. The primary filtering criterion utilized was the Settlement to Depth Ratio ($S_T/D_p < 1.0$). This constraint is grounded in the fundamental Principle of Conservation of Mass and the phase relationships of soil mechanics. According to classical consolidation theory (Terzaghi, 1943), settlement primarily occurs due to the expulsion of water and the reduction of the void ratio. However, the volume of soil solids (V_s) remains incompressible under engineering loads. Consequently, the total primary settlement (S_p), cannot mathematically equal or exceed the initial thickness of the compressible stratum (D_p), as this would imply a non-physical state where the void ratio becomes negative and the solid soil particles are eradicated [65]. During the parametric simulation, approximately 3,000 data points were identified where extreme combinations specifically maximum compression indices ($c_c = 3.15$) paired with high initial void ratios ($e_o = 15.0$) and shallow peat depths yielded settlement values exceeding the total stratum thickness. These anomalies represent the limits of linear analytical math when applied to ultra-soft organic soils. By performing Physically Informed Data Pruning and removing these up to 3,000 non-physical instances, the dataset was refined to 21,000 high-fidelity samples. This refinement is essential to eliminate model bias, ensuring that the ANN and SVR architectures learn the true mechanical boundary limits of tropical peat rather than mathematical errors. This optimized, physically constrained dataset serves as the definitive engine for subsequent model training and validation as per overall workflow of data cleaning as below in Fig 2.

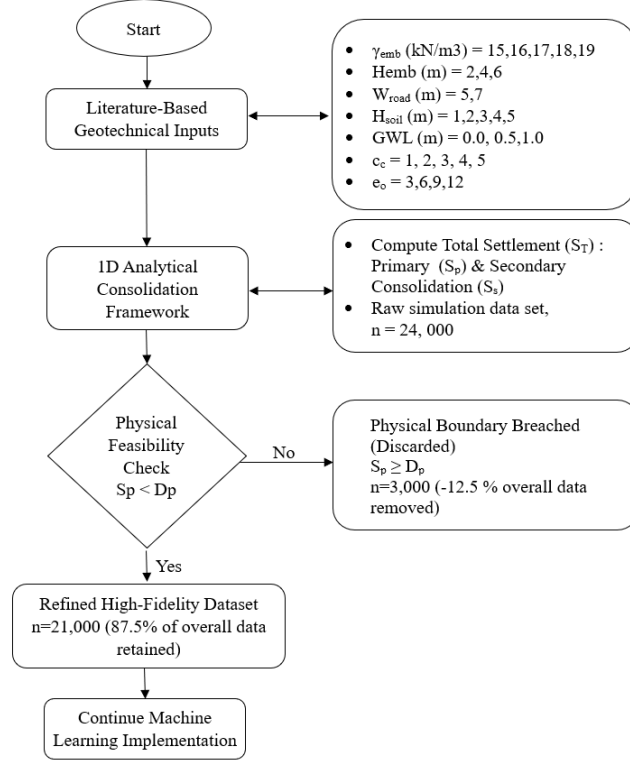


Fig. 2: Overall flowchart data refinement

2.3 Artificial Neural Network (ANN) Development

The ANN surrogate model was developed using a Multi-Layer Perceptron (MLP) architecture. The MLP is a feed-forward neural network that is well suited for modelling highly non-linear relationships between geotechnical input parameters and settlement response. In this study, the ANN was designed to predict embankment settlement over tropical peat by learning the complex consolidation behavior represented in the synthetic dataset, as illustrated in the methodology flowchart in Fig.3.

The ANN development process emphasized a balance between model complexity and generalization capability. This was important because an overly simple network may fail to capture the non-linear settlement behavior, while an overly complex network may overfit the training data and perform poorly on unseen cases. Therefore, the ANN architecture, activation function, optimizer, regularization method, and stopping criteria were carefully selected to improve predictive reliability.

The predictive engine was constructed using a three-tier neural network structure consisting of an input layer, hidden layers, and an output layer. The input layer consisted of nine neurons, representing the main embankment geometry and peat soil parameters used in the settlement prediction model. These input parameters were embankment unit weight (γ_{emb}), embankment height (H_{emb}), embankment width (B_1), unit weight of peat (γ_{peat}), saturated unit weight of peat ($\gamma_{saturated}$), groundwater level (GWL), peat depth (D_p), compression index (c_c), and initial void ratio (e_o).

A deep hidden-layer configuration was adopted, consisting of two hidden layers with 128 and 64 neurons, respectively. The first hidden layer with 128 neurons was used to capture broad and complex interactions among the input variables, while the second hidden layer with 64 neurons refined these extracted features into a more compact representation. This hierarchical structure allowed the ANN to learn latent patterns from the geotechnical inputs and map them to the corresponding settlement output.

The general mathematical form of the ANN model can be expressed as:

$$\hat{y} = f(X, W, b)$$

where $\hat{y}$ is the predicted settlement, X is the input feature vector, W represents the network weights, and b represents the bias terms. For each neuron, the output is calculated as:

$$z = \sum_{i=1}^n w_i x_i + b$$

$$\alpha = \phi(z)$$

where x_i is the input variable, w_i is the corresponding weight, b is the bias, z is the weighted summation, α is the neuron activation output, and ϕ is the activation function. The Hyperbolic Tangent (tanh) activation function was used in the hidden layers. Unlike the Rectified Linear Unit (ReLU), the tanh function produces a symmetric and zero-centered output ranging from -1 to +1. This characteristic is useful when working with normalized geotechnical datasets because it helps stabilize the learning process. The tanh activation function is given by:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

To ensure stable convergence and numerical robustness, the ANN was trained using the Adam optimizer, also known as Adaptive Moment Estimation. Adam was selected because it performs efficiently on non-convex optimization problems and automatically adjusts the learning rate during training based on first- and second-moment estimates of the gradients. The initial learning rate (eta) was set to 0.005.

The loss function was minimized during training by adjusting the network weights and biases. In this study, regularization was introduced using an L2 penalty, also known as weight decay, with a value of 0.0001. This penalty discourages excessively large weight values and helps produce a smoother and more physically reasonable settlement response surface. The regularized loss function can be written as:

$$L = MSE + \alpha \sum_{j=1}^m w_j^2$$

where L is the regularized loss, MSE is the mean squared error, α is the L2 regularization coefficient, and w_j represents the network weights.

The mean squared error component is defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual settlement value, $\hat{y}_i$ is the predicted settlement value, and n is the total number of observations.

The training process was bounded by a maximum of 3,000 epochs. However, to prevent over-memorization of the training dataset, an early-stopping mechanism was applied. A validation fraction of 10% was used to monitor the model's generalization performance during training. If the validation loss did not improve for 10 consecutive iterations, the training process was automatically terminated. This ensured that the final ANN model retained the best balance between accuracy and generalization.

Overall, the ANN surrogate model was developed using a structured MLP architecture, normalized geotechnical inputs, two hidden layers, (tanh) activation, Adam optimization, L2 regularization, and early stopping. These components allowed the model to capture the non-linear settlement behaviour of tropical peat while reducing the risk of overfitting.

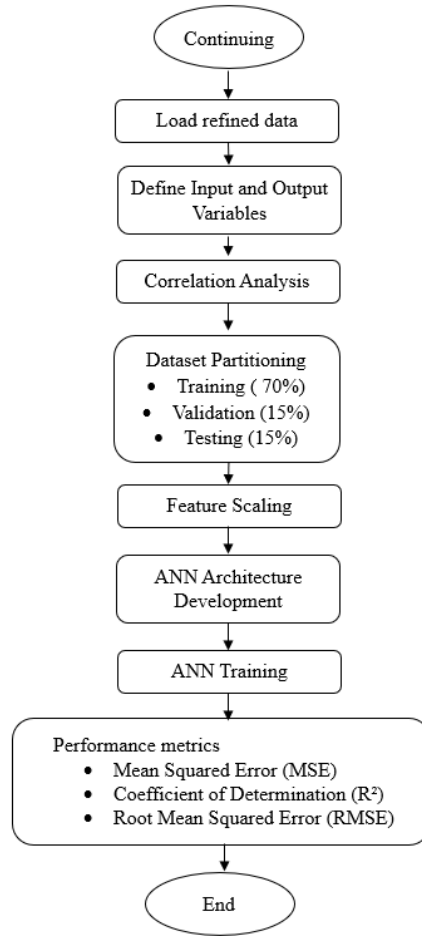


Fig.3 : Overall flowchart for ANN Architectural development

2.4 Support Vector Regression (SVR) Development

The second machine learning model developed for settlement prediction was SVR. The SVR model was implemented to provide a robust benchmark against the ANN architecture. SVR was selected because of its ability to model complex and non-linear geotechnical relationships while maintaining good generalization performance. This is achieved through the principle of Structural Risk Minimization (SRM), which aims to balance model complexity and prediction error, thereby reducing the risk of overfitting. To enable the SVR model to capture the non-linear consolidation behavior of Malaysian peat, a kernel mapping approach was employed. Since the dataset consisted of nine input parameters and involved highly non-linear settlement responses, the Radial Basis Function (RBF) kernel was selected. The RBF kernel maps the original input space into a higher-dimensional feature space, where a linear hyperplane can be used to approximate the settlement response surface more effectively. The RBF kernel is expressed as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$$

where $K(\mathbf{x}_i, \mathbf{x}_j)$ is the kernel function, $\mathbf{x}_i$ and $\mathbf{x}_j$ are input feature vectors, and γ is the kernel coefficient that controls the influence range of each training sample.

Unlike ANN models, where the network architecture is manually defined, SVR performance is strongly influenced by its internal hyperparameters. Therefore, automated hyperparameter tuning was carried out using Grid Search Cross-Validation (GridSearchCV). The GridSearchCV procedure was implemented within a consolidated machine learning pipeline to ensure that data preprocessing and model training were

performed correctly. In particular, the Standard Scaler was fitted only on the training folds during cross-validation, preventing data leakage from the validation folds into the training process.

The hyperparameter search space consisted of three main SVR parameters: the penalty factor (C), the epsilon-insensitive loss parameter (ϵ), and the kernel coefficient (γ). The penalty factor C controls the trade-off between maximizing the regression margin and minimizing the training error. In this study, values of $C = 1, 10, 100$, and 1000 were tested. A smaller C value allows a wider margin with greater tolerance for errors, while a larger C value forces the model to reduce training errors more strictly.

The epsilon-insensitive parameter ϵ defines the width of the tolerance zone within which prediction errors are ignored. Values of $\epsilon = 0.001, 0.01, 0.05$, and 0.1 were evaluated. A larger ϵ value produces a smoother regression function by ignoring small deviations, whereas a smaller value increases model sensitivity to minor prediction errors.

The kernel coefficient, γ controls how far the influence of a single training sample extends. The tested values were $\gamma = \text{scale}, 0.001, 0.01, 0.1$, and 1 . A small γ value results in a broader influence range, producing a smoother model, while a large γ value gives each sample a narrower influence range, which may lead to a more complex regression surface.

For model tuning and generalization assessment, 5-fold cross-validation ($CV = 5$) was employed during the GridSearchCV process. In this procedure, the training dataset was divided into five subsets. The model was trained using four subsets and validated using the remaining subsets, with the process repeated until each subset had served once as the validation fold. This approach ensured that the selected best estimator was not biased toward a particular portion of the synthetic peat dataset.

Overall, the optimized SVR model was developed using RBF kernel mapping, standardized input features, automated hyperparameter tuning, and cross-validation as shown in the Fig 4. This workflow provided a reliable and systematic approach for producing a generalized SVR surrogate model suitable for predicting tropical peat settlement behavior.

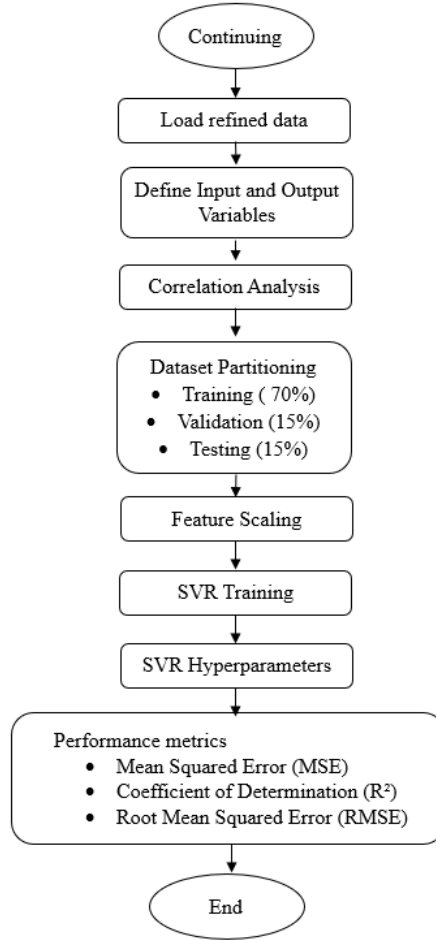


Fig. 4: Overall flowchart for SVR Architectural development

2.5 Performance Evaluation and Parametric Verification

The final phase of the methodology involved a head-to-head performance evaluation between the optimized ANN and SVR surrogate models. This stage was conducted to determine which model provided the most reliable and accurate prediction performance after optimization.

To ensure a comprehensive and unbiased assessment, the refined dataset was divided into three subsets: training, validation, and independent testing. The training subset, representing 70% of the total data, was used to develop the predictive relationship between the input parameters and the target output. The validation subset, representing 15% of the data, was used during model tuning to monitor generalization performance and reduce the risk of overfitting. The remaining 15% of the dataset was reserved as an independent testing subset.

The independent testing subset served as the final “unseen” dataset for evaluating model reliability. Since these data were not used during training or validation, the testing results provided a more realistic indication of how well each model could perform on new cases. The predictive performance of the ANN and SVR models was then compared using statistical error metrics, including the Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE).

In addition to statistical comparison, parametric verification was carried out to examine whether the optimized surrogate models could reproduce physically meaningful trends across the investigated parameter ranges. This verification step was important to confirm that the models were not only statistically accurate, but also consistent with the expected behavior of the system being studied. Overall, the model with higher predictive

accuracy, lower error values, and stronger parametric consistency was considered the more reliable surrogate model.

2.5.1 Statistical Metrics

The predictive accuracy of each model was evaluated using three widely used statistical performance indicators: the Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). These metrics were selected because they provide complementary information about model performance, including the strength of agreement between observed and predicted values, the magnitude of prediction errors, and the model's sensitivity to larger residuals. The Coefficient of Determination, R^2 , was used to measure how well the predicted values explained the variation in the observed data. A value of R^2 closer to 1 indicates stronger predictive performance, while a value closer to 0 indicates weaker agreement.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the observed values, and n is the total number of observations.

The Root Mean Square Error (RMSE) was applied to quantify the average magnitude of prediction error. RMSE gives greater weight to large errors because the residuals are squared before averaging. Therefore, it is useful for identifying models that produce large deviations from observed values. The Mean Absolute Error (MAE) was used to represent the average absolute difference between observed and predicted values. Unlike RMSE, MAE treats all errors equally and is less sensitive to extreme residuals. Since the error is expressed in the same unit as the measured variable, it provides a direct interpretation of the average physical error in meters.

Overall, a model with a higher R^2 value and lower RMSE and MAE values was considered to have better predictive accuracy. Using these three metrics together allowed for a more reliable assessment of model performance, as each metric highlights a different aspect of prediction quality.

2.5.2 Parametric Trend Reproduction

Critically, to move beyond simple statistical correlation, the models were subjected to a Parametric Trend Verification. This involved isolating individual inputs such as Peat Depth (D_p) or Compression Index (c_c) while holding others constant to ensure the Black Box predictions adhere to established soil mechanics principles. This verification ensures that the models are not only mathematically accurate but geotechnically sound, providing a reliable transition to the results and discussion presented in the following chapter.

3 Results and Discussion

3.1 Correlation Heatmap

Prior to deploying the predictive machine learning architectures, a Pearson correlation analysis was conducted to quantify the linear dependencies between the embankment

geometry, peat state parameters, and the target variable which is the Total Settlement S_T which shown in Fig.5 below. The resulting heatmap highlights that settlement over tropical peat is primarily governed by intrinsic material properties rather than external embankment geometry.

The strongest positive correlations with settlement were observed for peat depth, D_p which the value showed was 0.55, and the compression index, c_c was 0.53. From a geotechnical perspective, this trend confirms established soft soil mechanics as thicker peat layer provides a larger compressible volume subjected to embankment loading, while a higher c_c denotes a more highly compressible organic fabric. This dependency is perfectly aligned with the classical one-dimensional consolidation framework:

As dictated by this governing equation, settlement increases proportionally with both stratum thickness and compression index, validating the statistical outputs of the heatmap. A highly insightful observation is the moderate negative correlation of -0.30 between the initial void ratio e_o and settlement in which means higher initial e_o give smaller S_T . While high void ratios are typically symptomatic of highly compressible peat, the theoretical settlement equation isolates $(1 + e_o)$ in the denominator. Consequently, for a given c_c and stress increment, an isolated mathematical increase in e_o reduces the overall volumetric strain. This indicates that the dataset successfully uncouples the independent mechanical contributions of c_c and e_o , rather than treating them as a singular, blended indicator of peat fabric.

Furthermore, the state variables influencing initial effective stress bulk unit weight (γ_{soil}), saturated unit weight (γ_{sat}), and Groundwater Level (GWL) all exhibited weak negative correlations ranging from -0.10 to -0.12. In geotechnical practice, a deeper water table or higher soil unit weight increases the initial effective overburden stress (σ'_o). As σ'_o increases, the overall stress ratio $\left(\frac{\sigma'_o + \Delta\sigma}{\sigma'_o}\right)$ decreases, thereby reducing the calculated primary settlement. While numerically weak, these linear correlations are physically sound.

Embankment parameters showed surprisingly weak linear influence, with embankment height (H_{emb}) showing a correlation of only 0.14, and width (B_1) showing negligible impact. This does not imply that embankment geometry is geotechnically insignificant rather, it highlights that the propagation of applied stress ($\Delta\sigma$) through the peat stratum is governed by non-linear spatial distributions for example Boussinesq stress bulbs that simple pairwise Pearson correlation cannot fully capture.

Additionally, the matrix revealed a perfect collinearity 1.00 between γ_{soil} and γ_{sat} . While such multicollinearity causes mathematical instability in conventional multiple linear regression models, it justifies the necessity of the proposed study. Advanced machine learning architectures, specifically Artificial Neural Networks (ANN) and Support Vector Regression (SVR), are inherently resilient to collinear inputs through weight penalization and kernel mapping.

Ultimately, the correlation heatmap confirms that while the fundamental physical drivers of peat settlement (c_c and D_p) can be isolated linearly, the complex interplay of geometry, stress distribution, and void ratio requires the sophisticated, non-linear processing capabilities of the developed ML surrogates.

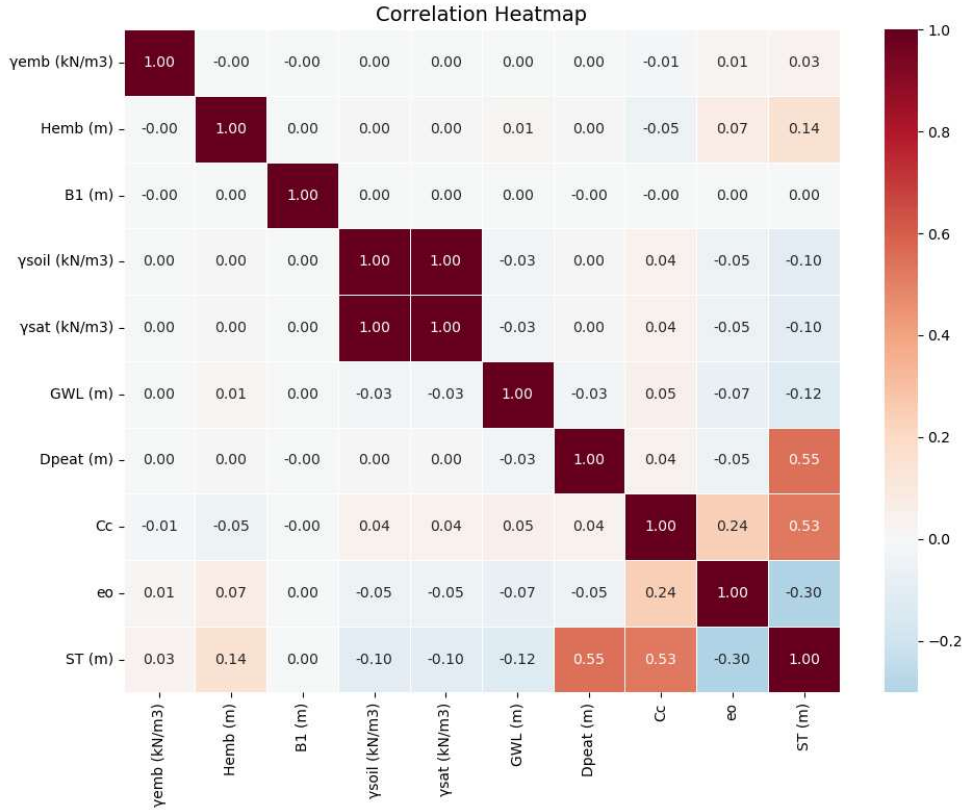


Fig.5 : Pearson's correlation Heatmap

3.2 Comparative Performance of Machine Learning Surrogates

The performance of the ANN and SVR models was evaluated using the coefficient of determination, R^2 , mean absolute error, MAE, and root mean square error, RMSE. The results are summarized in Table 6.

Table 6 ANN & SVR metric performance

Model	Dataset	R^2	MAE (m)	RMSE (m)
ANN	Training	0.999672	0.009584	0.013151
	Validation	0.999652	0.009942	0.013432
	Testing	0.999655	0.009646	0.013190
SVR	Training	0.999922	0.005516	0.006406
	Validation	0.999893	0.005894	0.007461
	Testing	0.999872	0.006025	0.008033

Both models demonstrated excellent predictive capability, with R^2 values greater than 0.999 across the training, validation, and testing datasets. This indicates that both ANN and SVR were able to capture the dominant settlement response patterns within the dataset. The close agreement between training, validation, and testing results also suggests strong generalization capability and limited evidence of overfitting.

For the ANN model, the testing R^2 was 0.999655, with MAE and RMSE values of 9.646 mm and 13.190 mm, respectively. These low error magnitudes indicate that the ANN successfully learned the nonlinear relationship between the nine input parameters and settlement. The similarity between the training, validation, and testing errors confirms

that the early-stopping and L2 regularization strategies were effective in controlling overfitting.

The SVR model produced even stronger predictive performance. On the independent testing set, SVR achieved an R^2 of 0.999872, with MAE and RMSE values of 6.025 mm and 8.033 mm, respectively. Compared with ANN, the SVR reduced the testing MAE by approximately 37.5% and the testing RMSE by approximately 39.1%. This demonstrates that the optimized SVR model provided more accurate settlement predictions than the ANN model for the present dataset.

The superior performance of SVR may be attributed to the suitability of the radial basis function kernel in modelling nonlinear geotechnical behaviour. The RBF kernel allows the SVR to map the original input variables into a higher-dimensional feature space, enabling it to capture complex settlement response surfaces with high precision. This is especially relevant for peat settlement, where the response is governed by nonlinear interactions between peat thickness, compressibility, void ratio, groundwater condition, and embankment loading.

Overall, the results indicate that the SVR model is the more reliable surrogate model for predicting settlement of embankments over tropical peat. While the ANN model also achieved excellent accuracy, the SVR provided consistently lower errors across all datasets. The heatmap further supports the modelling outcome by confirming that settlement is mainly controlled by D_P and C_c , while other parameters contribute through weaker or nonlinear interactions. Therefore, the use of nonlinear machine learning models is justified for this problem, as conventional linear interpretation alone would not fully capture the coupled behaviour of peat settlement.

3.3 Parametric Trends for High-Impact Features

The parametric verification was conducted to examine whether the ANN and SVR models could reproduce geotechnically meaningful settlement trends when key input variables were varied independently. Based on the correlation heatmap, D_P , c_c , and e_o , were identified as the most influential parameters affecting total settlement, S_T . Although groundwater level, GWL , embankment height, H_{emb} , and embankment unit weight, γ_{emb} , showed weaker linear correlations, they were also assessed because they directly influence the effective stress condition and applied embankment surcharge.

For the compression index, c_c , settlement increased significantly as c_c increased from 1.0 to 5.0 for all peat types. For fibric peat, the manual settlement increased from 0.402 m to 2.144 m. Hemic peat increased from 0.341 m to 1.793 m, while sapric peat increased from 0.301 m to 1.572 m. This confirms that c_c is one of the dominant parameters controlling peat settlement. The trend is physically reasonable because a higher compression index represents a more compressible soil skeleton, resulting in greater volume reduction under the same surcharge loading. Both ANN and SVR reproduced this increasing trend closely, confirming that the models captured the fundamental compressibility behaviour of peat as shown in the Fig 7.

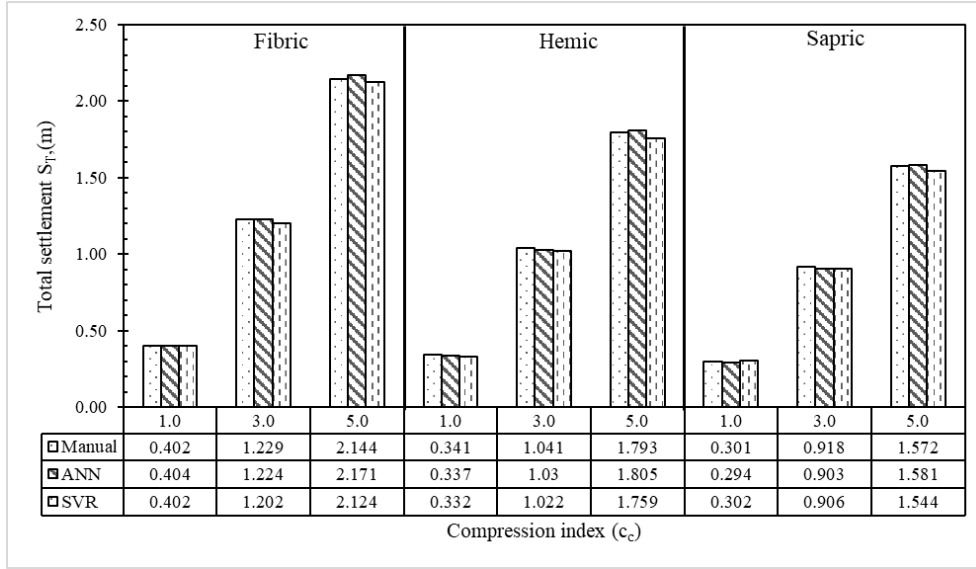


Fig.7: Parametric verification of the ANN and SVR models against the analytical baseline for effect of c_c variations on S_T across different peat decomposition states.

The void ratio, e_o , produced an inverse trend, where settlement decreased as e_o increased from 3.0 to 12.0 as shown in Fig.8. For fibric peat, settlement reduced from 3.396 m to 0.842 m. Hemic peat reduced from 2.589 m to 0.714 m, while sapric peat reduced from 2.220 m to 0.630 m. This trend is consistent with the consolidation equation because e_o appears in the denominator as $(1 + e_o)$. Therefore, when c_c and other stress conditions are kept constant, increasing e_o reduces the settlement term. This should be interpreted as a controlled parametric effect, not as a general field statement that higher void ratio always produces lower compressibility, since natural peat behavior depends on the coupled relationship between e_o , c_c , organic content, fibre structure, and stress history.

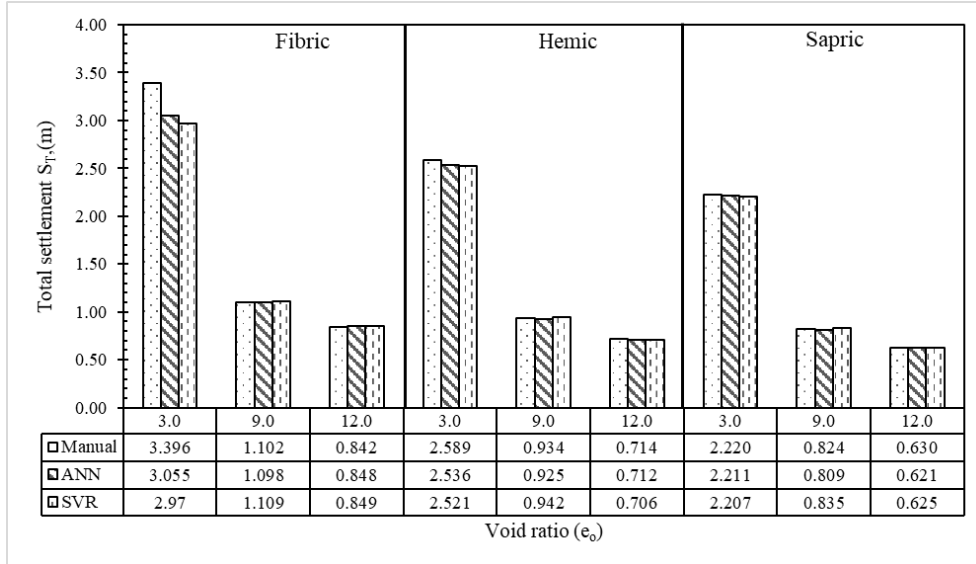


Fig.8: Parametric verification of the ANN and SVR models against the analytical baseline for effect of e_o variations on S_T across different peat decomposition states.

Peat depth, D_P , also showed a strong positive influence on settlement. As D_P increased from 1.0 m to 5.0 m, fibric peat settlement increased from 0.447 m to 1.893 m, hemic peat from 0.416 m to 1.509 m, and sapric peat from 0.390 m to 1.283 m. This agrees with the heatmap, where D_P had the strongest positive correlation with S_T .

Geotechnically, a thicker peat deposit provides a larger compressible volume, leading to greater total vertical deformation. The ANN and SVR predictions followed the manual trend closely, indicating that both models preserved the expected proportional effect of peat thickness as shown in the Fig.9.

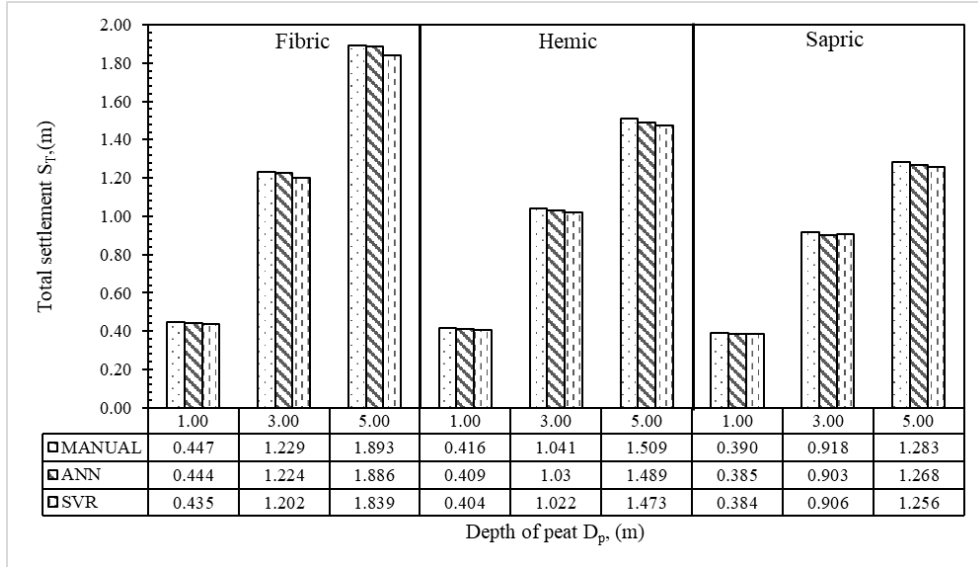


Fig.9 : Parametric verification of the ANN and SVR models against the analytical baseline for effect of D_p variations on S_T across different peat decomposition states.

Groundwater level showed a decreasing settlement trend within the adopted variable definition. As GWL increased from 0.0 m to 1.0 m, settlement decreased from 1.697 m to 1.022 m for fibric peat, from 1.263 m to 0.905 m for hemic peat, and from 1.062 m to 0.818 m for sapric peat. This supports the weak negative correlation observed in the heatmap. The response indicates that groundwater condition influences settlement through changes in effective stress and stress ratio, although its effect is less dominant than c_c and D_p . Both machine learning models reproduced the decreasing trend without generating physically inconsistent predictions as produced by the Fig.10 below.

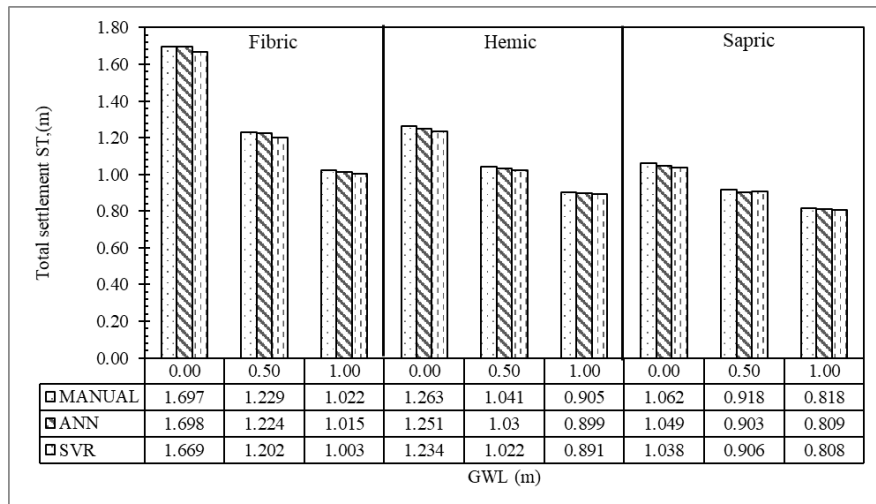


Fig.10: Parametric verification of the ANN and SVR models against the analytical baseline for effect of GWL variations on S_T across different peat decomposition states.

For embankment height, H_{emb} , settlement increased as H_{emb} increased from 2.0 m to 6.0 m. The manual settlement increased from 0.946 m to 1.403 m for fibric peat, from 0.775 m to 1.208 m for hemic peat, and from 0.666 m to 1.080 m for sapric peat. This is

expected because a higher embankment produces a larger surcharge stress, $\Delta\sigma$, and therefore a greater consolidation response. However, the increase was less pronounced than that caused by c_c and D_p , confirming that peat compressibility and thickness are more dominant than embankment geometry alone as shown in the Fig.11 below.

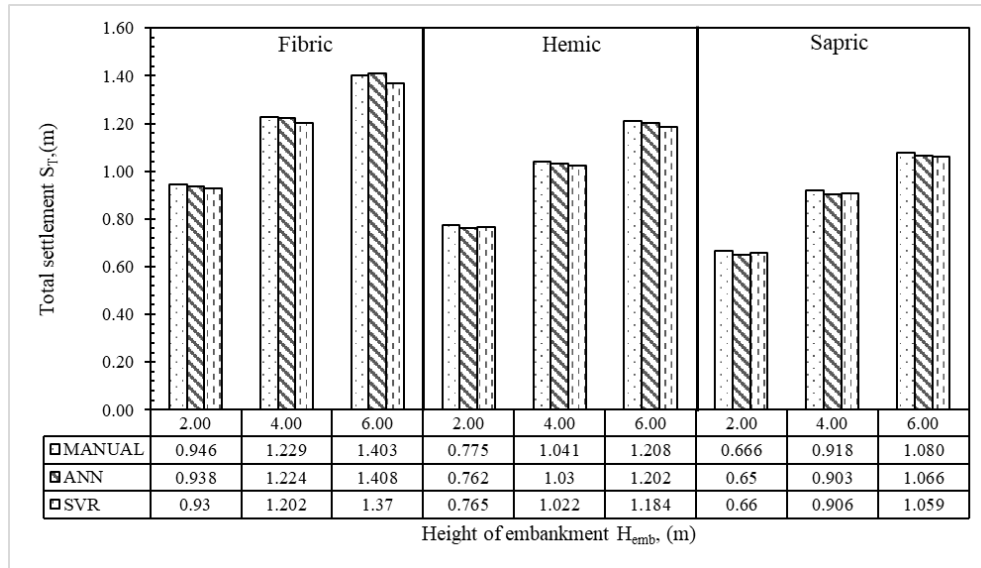


Fig.11 : Parametric verification of the ANN and SVR models against the analytical baseline for effect of H_{emb} variations on S_T across different peat decomposition states.

The effect of embankment unit weight, γ_{emb} , was comparatively smaller. Increasing γ_{emb} from 15 kN/m^3 to 19 kN/m^3 increased settlement from 1.177 m to 1.276 m for fibric peat, from 0.991 m to 1.086 m for hemic peat, and from 0.871 m to 0.961 m for sapric peat. This modest increase explains why γ_{emb} showed only a weak correlation with ST in the heatmap. Nevertheless, the monotonic increase confirms that the models retained the correct geotechnical response to increasing embankment load intensity as shown in the Fig. 12 below.

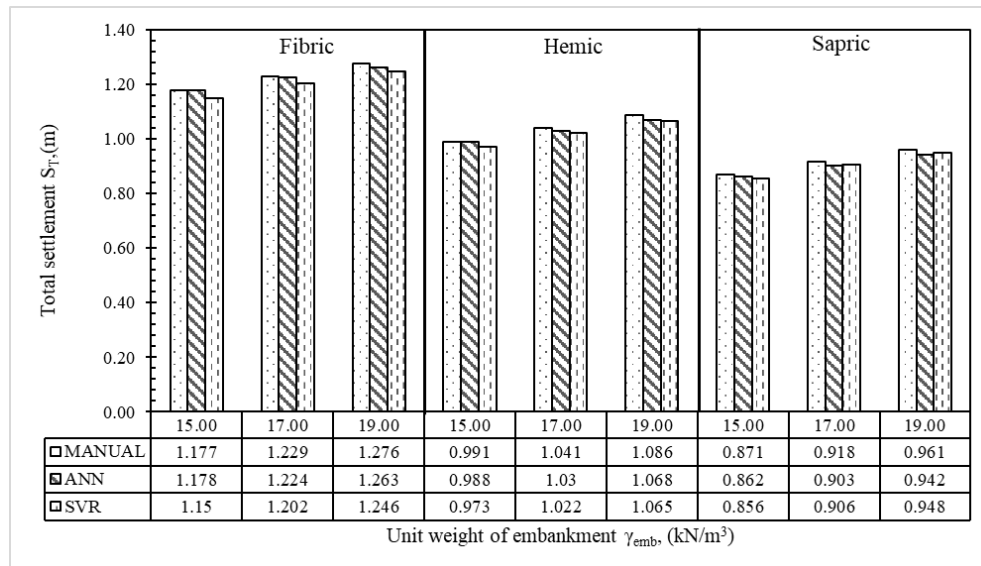


Fig.12: Parametric verification of the ANN and SVR models against the analytical baseline for effect of γ_{emb} variations on S_T across different peat decomposition states.

Across all parametric cases, fibric peat consistently produced the highest settlement, followed by hemic and sapric peat. This ranking reflects the progressive reduction in

compressibility with increasing peat decomposition. Fibric peat generally contains a more open and highly compressible organic structure, while sapric peat is more decomposed and comparatively less deformable.

Overall, both ANN and SVR successfully reproduced the manual parametric trends. The ANN showed slightly closer agreement in these selected parametric plots, with an average absolute deviation of approximately 0.017 m, while SVR showed approximately 0.027 m. However, the independent testing results showed that SVR achieved lower overall MAE and RMSE. Therefore, both models can be considered physically reliable, with SVR providing stronger global predictive accuracy and ANN showing very close agreement in the plotted parametric verification cases. The results confirm that the developed surrogate models are not merely statistically accurate, but also geotechnically consistent.

3.4 Engineering Utility: Statistical Accuracy versus Physical

Consistency

The comparative evaluation of the ANN and SVR models highlights a critical distinction in machine learning applications for geotechnical engineering, namely the trade-off between global statistical accuracy and physical engineering reliability.

Based strictly on global error minimization, the SVR model achieved the best predictive performance. As established in Section 3.2, the SVR yielded the lowest testing RMSE of 8.033 mm and MAE of 6.025 mm. In comparison, the ANN produced a testing RMSE of 13.190 mm and MAE of 9.646 mm. This statistical superiority indicates that the SVR, driven by its optimized kernel function, is highly effective for predicting settlement values located within the same well-defined statistical domain as the training data.

However, geotechnical design does not depend solely on global error minimization. A reliable surrogate model must also reproduce physically meaningful trends when key design parameters are varied independently. The parametric verification in Section 3.3 demonstrated that the ANN followed the manual consolidation-based trends more closely than the SVR. During these isolated sensitivity analyses, the ANN produced an average absolute deviation of approximately 0.017 m from the manual analytical baseline, compared with 0.027 m for the SVR. This indicates that the ANN better preserved the expected physical effects of compression index (c_c), peat thickness (D_p), initial void ratio (e_o), groundwater level (GWL), embankment height (H_{emb}), and embankment unit weight (γ_{emb}).

Therefore, the selection of the appropriate surrogate model depends on the specific engineering objective. The SVR is recommended for direct high-precision settlement prediction when the site-specific input data are fully defined and remain within the established training domain. In contrast, for geotechnical design applications involving sensitivity analysis, parametric interpretation, and assessment of soil behaviour under changing boundary conditions, the ANN is considered the more reliable engineering surrogate. Although the SVR produced lower global RMSE and MAE values, the ANN demonstrated stronger physical fidelity and greater consistency with classical consolidation theory, making it the more geotechnically robust tool for practical decision-making and preliminary design.

4 Conclusions

This study developed and evaluated two machine learning surrogate models, namely Artificial Neural Network (ANN) and Support Vector Regression (SVR), for predicting embankment settlement over tropical peat. The models were trained and tested using key embankment geometry, groundwater, and peat soil parameters, including embankment unit weight, embankment height, embankment width, soil unit weight, saturated unit weight, groundwater level, peat depth, compression index, and initial void ratio.

The correlation analysis showed that settlement was mainly governed by peat depth and compression index, which recorded the strongest positive relationships with total settlement. This confirms that peat thickness and compressibility are the dominant factors controlling settlement behaviour. Initial void ratio showed a negative relationship with settlement under the controlled parametric condition, while embankment height, groundwater level, and embankment unit weight showed weaker but physically meaningful influences.

Both ANN and SVR achieved excellent predictive performance, with R^2 values greater than 0.999 for the training, validation, and testing datasets. The SVR model produced the best global statistical accuracy, achieving a testing RMSE of 8.033 mm and MAE of 6.025 mm. In comparison, the ANN model achieved a testing RMSE of 13.190 mm and MAE of 9.646 mm. These results indicate that SVR is highly effective for direct settlement prediction within the established training domain.

However, the parametric verification demonstrated that the ANN model followed the manual consolidation-based trends more closely than the SVR model. The ANN produced an average absolute deviation of approximately 0.017 m from the analytical baseline, compared with 0.027 m for the SVR. This indicates that ANN provided stronger physical consistency when individual geotechnical parameters were varied independently.

Therefore, while SVR is statistically superior in terms of global prediction accuracy, ANN is considered the more geotechnically robust surrogate for engineering design applications requiring sensitivity analysis, physical interpretation, and consistency with classical consolidation behaviour. The findings demonstrate that machine learning models should not be evaluated based on error metrics alone, but also through parametric verification to ensure that predictions remain physically meaningful for practical geotechnical decision-making.

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Declarations

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Conflict of interest/Competing interests Not Applicable

Ethics approval and consent to participate Not Applicable

Consent for publication Not Applicable

Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. The study generated a synthetic database of 24,000 scenarios, which was refined to 21,000 high-fidelity samples used for model training

Materials availability Not Applicable

Code availability The code for the developed Artificial Neural Network (ANN) and Support Vector Regression (SVR) models is available from the corresponding author upon reasonable request. The study focused on developing and evaluating these machine learning surrogates for settlement prediction

Author contribution Not Applicable

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